

Python : 3.14.7  
NumPy : 2.5.2  
SciPy : 1.18.0  
Matplotlib : 3.11.1

mean of  $\sin(x)$  over one period : 0.0

integral of  $\sin(x)$  from 0 to  $\pi$  : 2.0 (exact value 2)

figure written to figures/lab00\_check.png

PLATFORM READY.